

Event-Aligned Singular Value Decomposition

for Data-Driven Pattern Kernel Discovery in Cryptocurrency Perpetual Futures

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Thesis

Discover predictive multi-candle price patterns without hand-designing them. Define events *analytically* (so they stay interpretable), extract aligned OHLCV windows, decompose into four signal channels, let SVD find the dominant *shapes*, then gate on forward-return correlation with FDR control. Inherits the interpretability of classical TA and the statistical efficiency of learned bases.

Pipeline

- Six analytical event types** — breakout bull/bear, turtle-soup false breakout bull/bear, trap bull/bear. Analytically defined, not learned $\Rightarrow$ interpretable.
- Four channels** per window: body, upper wick, lower wick, volume.
- Event-aligned matrix** $\rightarrow$ **SVD** $\rightarrow$ dominant shape variations.
- IC gate** with Benjamini–Hochberg FDR ($q = 0.05$) on forward returns.
- Online scoring** by cosine similarity against the surviving kernels.

The structural safeguard (the argument to a supervisor)

SVD operates on price geometry alone — it never sees forward returns. Basis discovery and label-fitting are separated by construction, so the usual “the shape was fit to the outcome” objection does not apply. Returns enter only at the gating step, where FDR is applied. That separation is the methodological point of the paper.

Data

Asset / venue	BTC-USD perp, Hyperliquid
Ticks	7,235,623 at 100 ms
Candles	12,059 at 60 s
Params	$W=20, N=20, K=3, \alpha=1.0, N_{\text{atr}}=14$
Events	1,775 total (147.2 per 1000 candles)

Breakouts most frequent (bear 522 > bull 418); traps rarest (136/172) — consistent with stricter multi-bar confirmation. All types clear $n_{\text{min}} = 100$ for stable SVD.

Result: 6 of 158 candidates survive FDR

Turtle bull, body, $k=0$	IC = +0.203
Turtle bull, body, $k=6$	IC = -0.201
Turtle bear, lower wick, $k=0$	IC = -0.262
Trap bull, upper wick, $k=6$	IC = +0.357
Trap bull, volume, $k=0$	IC = -0.270
Trap bear, body, $k=7$	IC = -0.272

No breakout kernel survives — despite high explained variance. The most-traded pattern in retail TA carries no forward-return information after multiple-testing correction. Turtle-soup and traps dominate; $|\text{IC}| \in [0.20, 0.36]$. All 6 have max cosine

similarity < 0.5 against each other (genuinely distinct shapes, not one kernel six ways).

Rolling-SVD stability

$\bar{\rho}$ over rolling windows: trap bull 0.941, turtle bull 0.920, breakout bear 0.855 (stable); turtle bear 0.719 (borderline); breakout bull 0.692, trap bear 0.669 (unstable). **The event types with the strongest IC are also the most temporally stable** — encouraging, and not something the method was tuned to produce.

The caveat that matters most

[!] **No kernel meets the robustness threshold** on the IS/OOS decay test. OOS ICs collapse: breakout bull $+0.243 \rightarrow -0.010$; trap bull upper wick $-0.474 \rightarrow +0.126$ (*sign flip*); trap bull volume $-0.449 \rightarrow +0.145$ (*sign flip*). Best decay ratios are turtle-type at 0.26–0.38, still under threshold. So: 6 kernels are *statistically significant in-sample after FDR*, and *none* is demonstrated to generalise. Do not let the abstract’s “6 significant kernels” be read as 6 usable signals — the paper is explicit; a skimming reader will not be.

Open decision (standing)

Publish as-is now (it is complete) vs. expand with the Spanning spectral sections first. **Recommendation on record: publish now, Spanning as paper #2.** P1 = polish to camera-ready, P2 = SSRN, P3 = arXiv q-fin.TR endorsement.

Validation framework

Walk-forward: **discovery** on $t \leq T_{\text{train}}$, **validation** strictly on $t > T_{\text{train}}$ — the split is what produced the negative robustness result above rather than hiding it. **Rolling-SVD stability** re-estimates the basis on moving windows and correlates successive bases ($\bar{\rho}$), which tests whether a “shape” is a real recurring geometry or a window-specific artifact. A kernel can be significant and unstable; both are reported.

Analytical basis dictionary

Discovered kernels are matched against closed-form families — polynomial, cubic, exponential ($a \in [3, 8]$) — plus an alignment diagnostic. Purpose: say *what shape was found*, in words, rather than shipping an opaque 20-dim vector. This is the interpretability half of the pitch and the part a classical-TA reader can check.

Why it suits a maths/signal-processing supervisor

Matched filtering + event-aligned averaging is the FINDCHIRP / gravitational-wave template-bank idea transplanted to price geometry; the basis is learned by SVD rather than derived from a physical model, and the multiple-testing control is what makes the transplant honest. That framing is the pitch for Bölcskei (signal-processing theory) and Teichmann (ML in finance).